

Breaking the Limit of Graph Neural Networks by Improving the Assortativity of Graphs with Local Mixing Patterns

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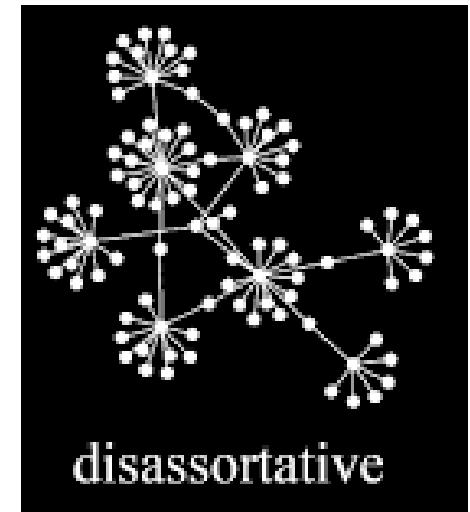
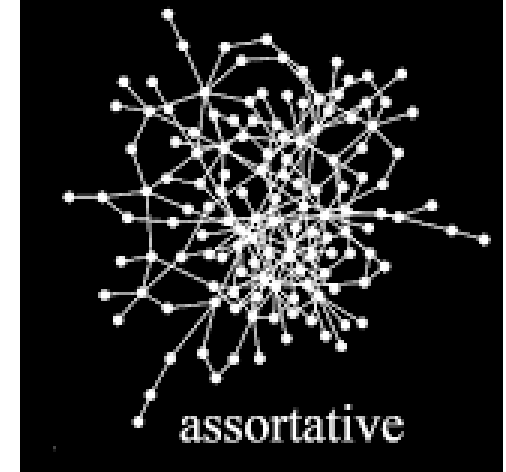
Assortative mixing

- Assortativity
- 동질성(동류성) : 비슷한 특징을 가진 노드들끼리 연결되는 성질
- Ex) social network – 같은 취미, 함께 아는 사람이 많음
citation network – 비슷한 분야의 논문들끼리 서로 인용함



Disassortative mixing

- Disassortativity
- 비동질성(비동류성) : 서로 다른 특징을 가진 노드들끼리 연결되는 성질
- Ex) heterosexual dating network – 다른 성별끼리 연결이 많음
ecological food webs where predator and prey – 포식자와 피식자 간 연결이 많음



Assortative mixing

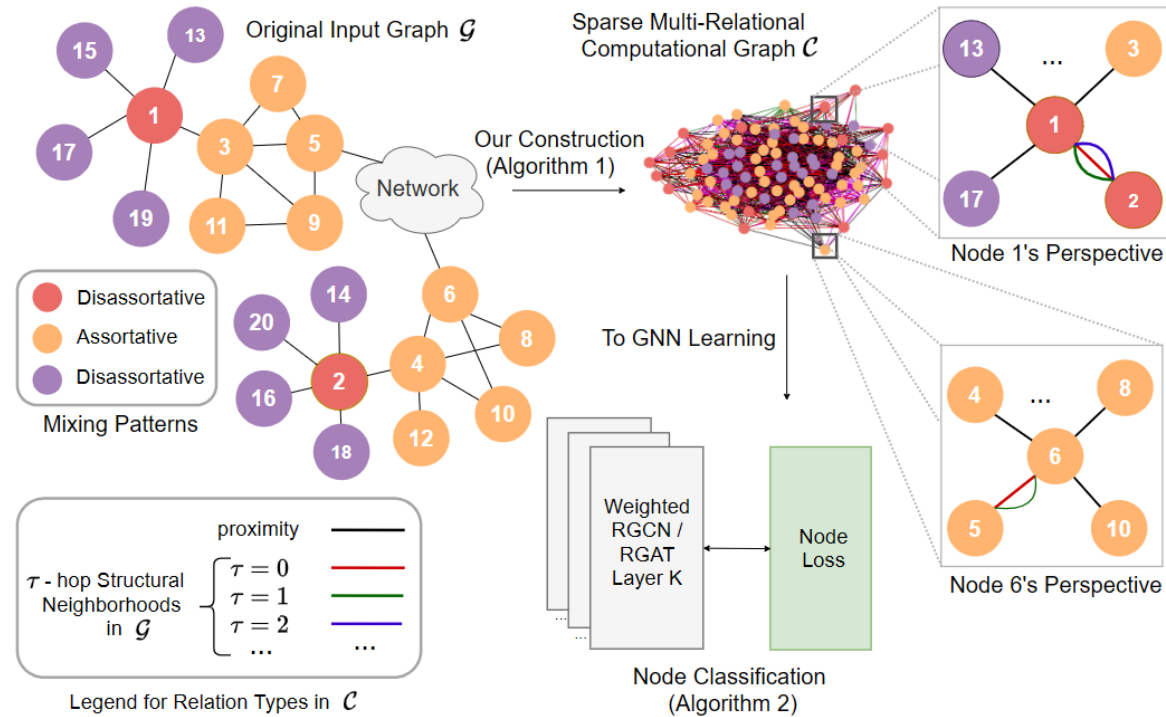


Figure 1: An input graph with diverse mixing pattern. Our pipeline uses both proximity and structural information to build a computation graph on which a GNN is run.

Preliminaries

- Global assortativity coefficient
 - To measure Mixing in Networks (비슷한 attribute/label을 가진 노드들끼리 연결되는 경향)

$$r_{\text{global}} = \frac{\sum_g M_{gg} - \sum_g a_g b_g}{1 - \sum_g a_g b_g}$$

$\sum_g a_g$: G에서 나가는 모든 노드들의 차수
 $\sum_g b_g$: G로 들어오는 모든 노드들의 차수

- Mixing matrix M

$$M_{gh} = \frac{1}{2m} \sum_{i:\tau_i=g} \sum_{j:\tau_j=h} A_{ij}$$

GNNs and Local Mixing

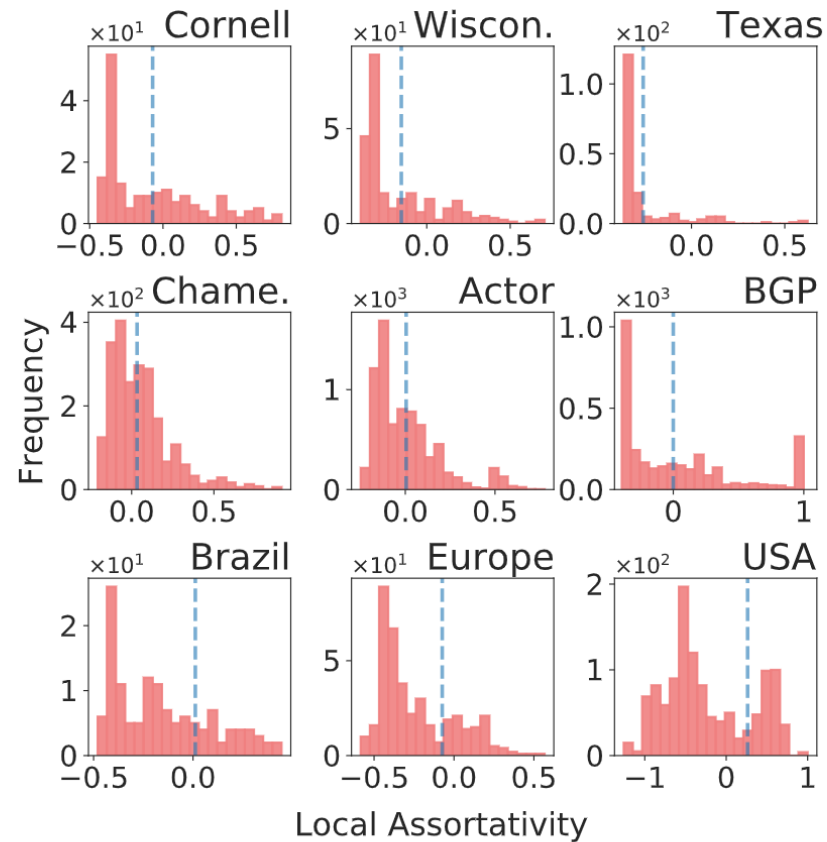


Figure 2: Observed distribution of node level assortativity in various graphs. The blue dotted line indicates global assortativity coefficient.

Computation Graph

- 노드가 얼마나 비슷한 연결을 가지는가

$$f_{\tau}(g, h) = f_{\tau-1}(g, h) + \mathcal{D}(s(\mathcal{N}_{\tau}(g)), s(\mathcal{N}_{\tau}(h)))$$

hop

거리

이웃 차수
sequence

Message Passing - WRGCN

- Aggregate function

$$m_u = \sum_{\tau \in \mathcal{R}} \sum_{v \in \mathcal{N}_1^\tau(u)} W_\tau h_v w_\tau(u, v) \alpha_\tau(u, v)$$

근접성+구조적 유사성

- Attention score

$$e_{u,v}^\tau = a_\tau(W_\tau h_u, W_\tau h_v) = a^T [W_\tau h_u || W_\tau h_v]$$

$$\alpha_\tau(u, v) = \frac{\exp(\text{LeakyReLU}(e_{u,v}^\tau))}{\sum_{x \in \mathcal{N}_1^\tau(u)} \exp(\text{LeakyReLU}(e_{u,x}^\tau))}$$

- Update function

$$h'_u = \sigma(W_{\text{self}} h_u + W_{\text{neig}} m_u)$$

Local Assortativity Distribution Shift

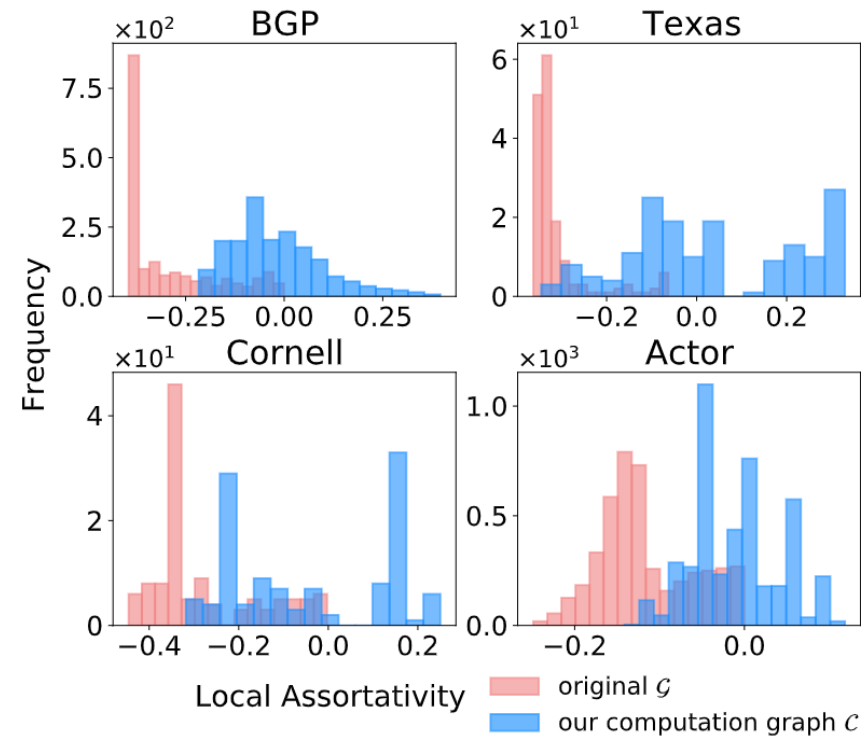
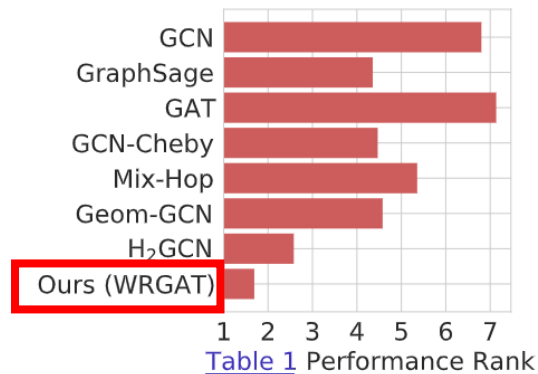


Figure 4: Distributions of node level assortativity for the original graph and the newly constructed computation graph on different datasets.

Performance of Node Classification

Table 1: Semi-supervised node classification showing mean test accuracy $\pm$ std. over 10 runs. Club Suit [♣] denotes result obtained from the best model variant of respective papers.

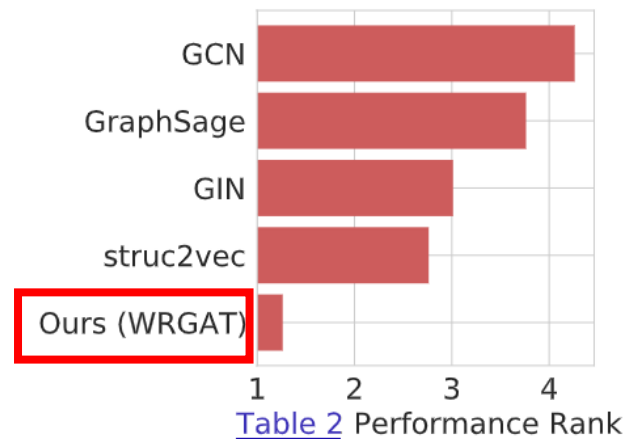
	Chameleon	Squirrel	Actor	Cornell	Texas	Wisconsin	Cora	Citeseer	Pubmed
GCN	59.82 $\pm$ 2.58	36.89 $\pm$ 1.34	30.26 $\pm$ 0.79	57.03 $\pm$ 4.67	59.46 $\pm$ 5.25	59.80 $\pm$ 6.99	87.28 $\pm$ 1.26	76.68 $\pm$ 1.64	87.38 $\pm$ 0.66
GraphSage	58.73 $\pm$ 1.68	41.61 $\pm$ 0.74	34.23 $\pm$ 0.99	75.95 $\pm$ 5.01	82.43 $\pm$ 6.14	81.18 $\pm$ 5.56	86.90 $\pm$ 1.04	76.04 $\pm$ 1.30	88.45 $\pm$ 0.50
GAT	54.69 $\pm$ 1.95	30.62 $\pm$ 2.11	26.28 $\pm$ 1.73	58.92 $\pm$ 3.32	58.38 $\pm$ 4.45	55.29 $\pm$ 8.71	86.37 $\pm$ 1.69	75.46 $\pm$ 1.72	87.62 $\pm$ 0.42
GCN-Cheby	55.24 $\pm$ 2.76	43.86 $\pm$ 1.64	34.11 $\pm$ 1.09	74.32 $\pm$ 7.46	77.30 $\pm$ 4.07	79.41 $\pm$ 4.46	86.86 $\pm$ 0.96	76.25 $\pm$ 1.76	88.08 $\pm$ 0.52
MixHop	60.50 $\pm$ 2.53	43.80 $\pm$ 1.48	32.22 $\pm$ 2.34	73.51 $\pm$ 6.34	77.84 $\pm$ 7.73	75.88 $\pm$ 4.90	83.10 $\pm$ 2.03	70.75 $\pm$ 2.95	80.75 $\pm$ 2.29
Geom-GCN ♣	60.90	38.14	31.63	60.81	67.57	64.12	85.27	77.99	90.05
H ₂ GCN ♣	59.39 $\pm$ 1.98	37.90 $\pm$ 2.02	35.86 $\pm$ 1.03	82.16 $\pm$ 4.80	84.86 $\pm$ 6.77	86.67 $\pm$ 4.69	87.67 $\pm$ 1.42	76.72 $\pm$ 1.50	88.50 $\pm$ 0.64
Ours (WRGAT)	65.24 $\pm$ 0.87	48.85 $\pm$ 0.78	36.53 $\pm$ 0.77	81.62 $\pm$ 3.90	83.62 $\pm$ 5.50	86.98 $\pm$ 3.78	88.20 $\pm$ 2.26	76.81 $\pm$ 1.89	88.52 $\pm$ 0.92



Performance of Node Classification

Table 2: Node classification on Air Traffic Networks and BGP Network. Mean test accuracy $\pm$ std. is shown over 20 runs.

	Brazil	Europe	USA	BGP
GCN	64.55 $\pm$ 4.18	54.83 $\pm$ 2.69	56.58 $\pm$ 1.11	53.33 $\pm$ 0.18
GraphSage	70.65 $\pm$ 5.33	56.29 $\pm$ 3.21	50.85 $\pm$ 2.83	65.19 $\pm$ 0.28
GIN	71.89 $\pm$ 3.60	57.05 $\pm$ 4.08	58.87 $\pm$ 2.12	49.51 $\pm$ 1.52
Struc2vec	70.88 $\pm$ 4.26	57.94 $\pm$ 4.01	61.92 $\pm$ 2.61	48.40 $\pm$ 1.39
Ours	76.92 $\pm$ 5.45	57.12 $\pm$ 2.81	63.02 $\pm$ 1.87	66.54 $\pm$ 0.48



Sensitivity Analysis

Analysis for disassortative networks

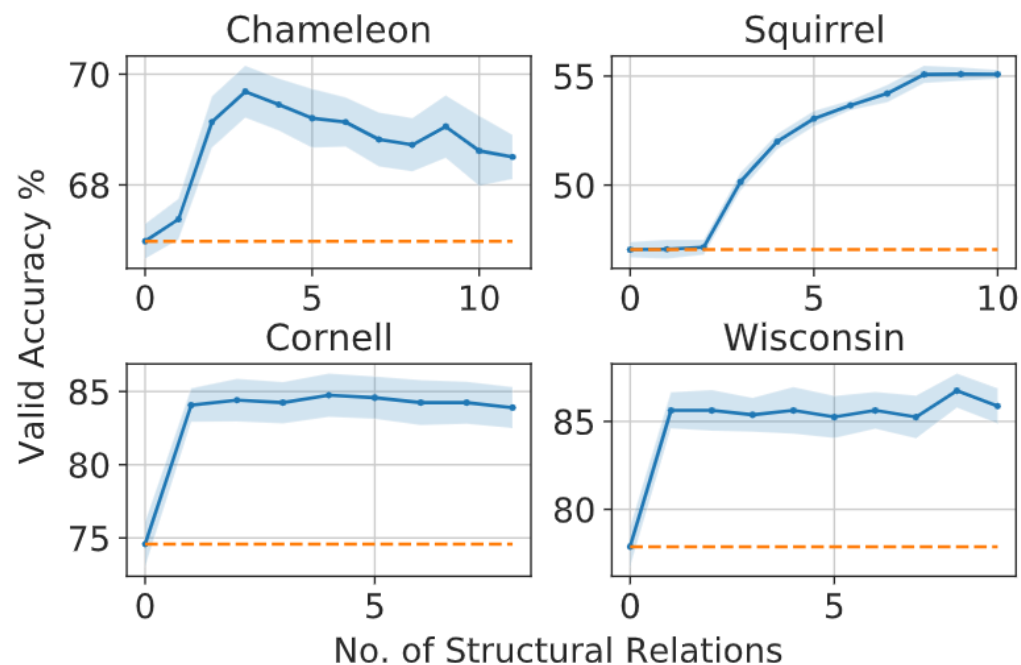


Figure 6: Sensitivity w.r.t structure relations. Baseline is using only proximity information.

Analysis for assortative networks

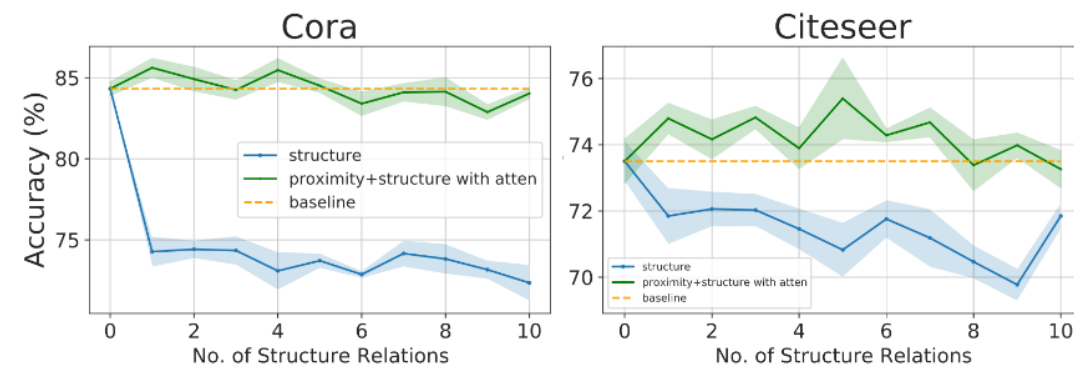


Figure 7: Sensitivity w.r.t structure relations. Baseline is using only proximity information.

Ablation Analysis

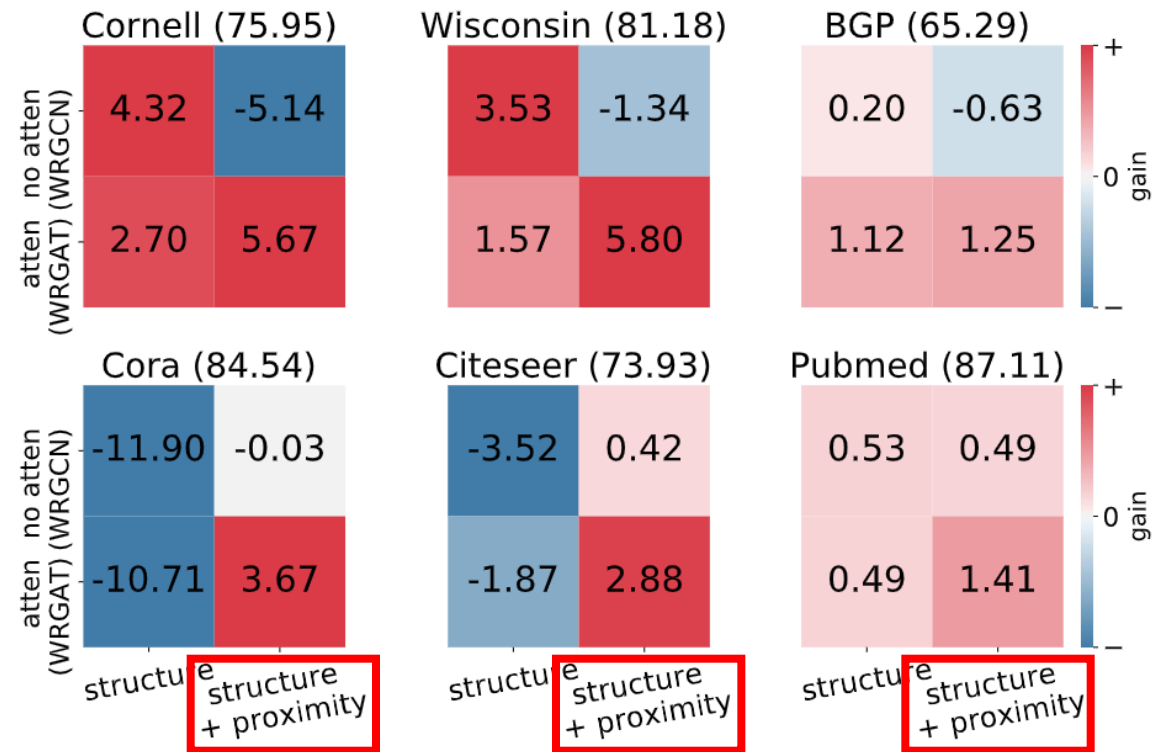


Figure 8: Ablation w.r.t different model and graph information choices. Baseline (proximity only) is shown in parenthesis. Numbers indicate gain over baseline.

Ablation Analysis

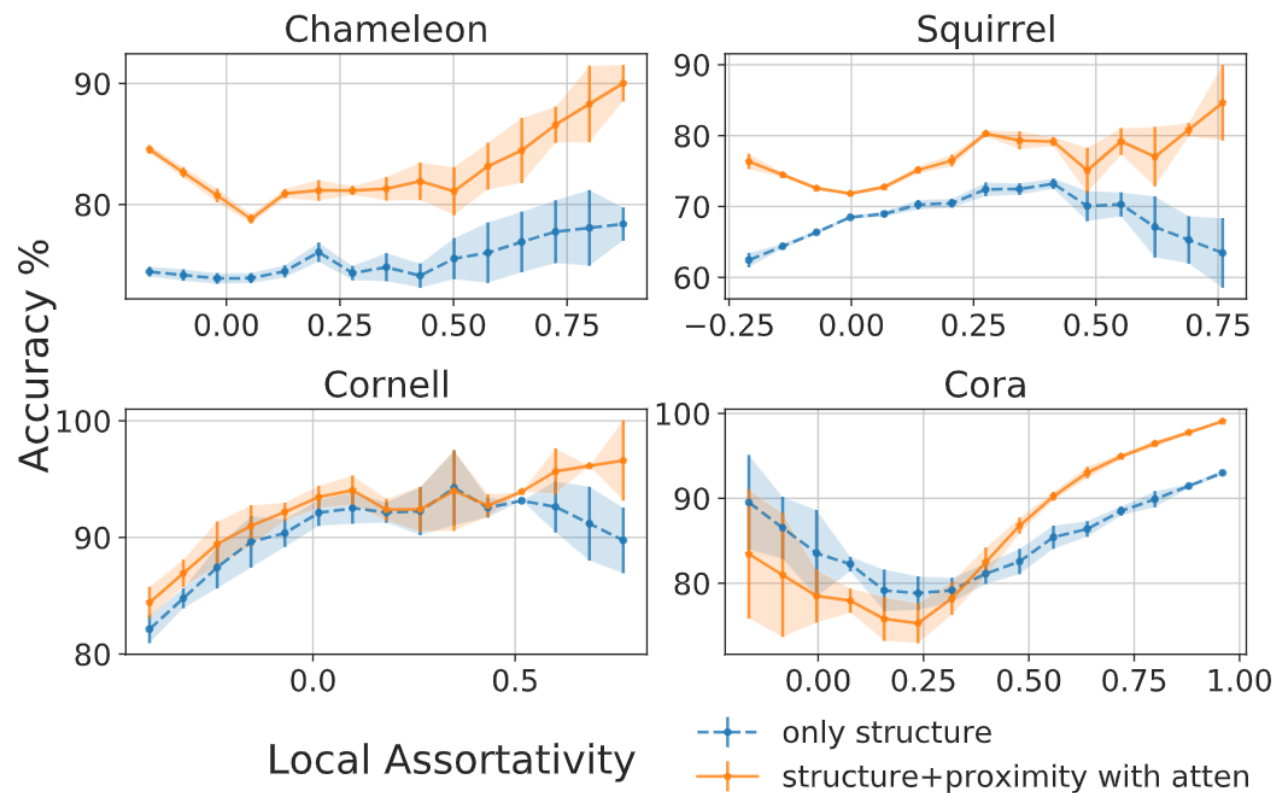


Figure 9: Ablation w.r.t graph info and local assortativity

Local Mixing Patterns

- GNN의 한계에 대한 새로운 관점
- 구조적 정보를 동질성에 이용하여 모델 성능 향상

